

Deep Variational Quantum Circuits with Barren-Plateau-Free Architectures

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ARTICLE INFO

Keywords:

variational quantum algorithms;
quantum machine learning; barren
plateaus

Citation:

K. N. Zhang, M.-H. Hsieh and D. C. Tao, "Deep variational quantum circuits with barren-plateau-free architectures," *Artif. Intel. Sci. Eng.*, vol. 2, no 1, pp. 66-84, Mar. 2026, doi: 10.23919/AISE.2026.000005

Publishing Editor:

Jun Li (Email: xtt@swu.edu.cn)

ABSTRACT

Variational quantum algorithms (VQAs) with random structures have poor trainability due to the exponentially vanishing gradient as the circuit depth and the qubit number increase. This result leads to a general belief that a deep circuit will not be feasible. In this work, we provide a viable solution to the vanishing gradient problem for deep VQAs with theoretical guarantees. Specifically, we prove that for quantum controlled-layer and quantum residual network (QResNet), architectures, the expectation of the gradient norm can be lower bounded by a value that is independent of the qubit number and the circuit depth. Our results follow from a careful analysis of the gradient behavior on parameter space consisting of rotation angles, as employed in almost all VQAs, instead of relying on impractical 2-design assumptions. We conduct several numerical experiments as verifications, where only our circuits are trainable and converge, while hardware-efficient and random circuits with similar number of parameters in comparison cannot converge.

1 Introduction

Recent experimental achievements have brought quantum advantage [1], [2] to reality on specific tasks. These developments stimulate quantum algorithm designs for the noisy intermediate-scale quantum (NISQ) era [3]. Variational quantum algorithms (VQAs) [4], [5] have mild requirements on the gate noise and the circuit connectivity, which makes them one of the most promising candidates for achieving practical quantum advantages on NISQ devices. Various VQAs have been explored in quantum machine learning [4], [5], [6], [7], [8], [9], [10], [11], [12], quantum chemistry [13], [14], and combinatorial optimizations [15], [16].

Despite the opportunities across applications, VQAs face a central bottleneck: trainability. As the qubit number N and circuit depth D grow, optimi-

zation can become difficult. A canonical manifestation is the barren plateau (BP) phenomenon [17], [18]: for random quantum circuits, objective values and gradients concentrate sharply, and gradients can decay exponentially with the qubit number.

In practice, BPs impede training in two complementary ways. ① First-order optimization methods such as gradient descent (GD) may require an exponential number of iterations to make meaningful progress [19]. ② Accurately estimating the gradient direction can demand an exponential number of measurement shots due to concentration and finite-sampling fluctuations [20]. Subsequent works established that BP is not limited to globally random circuits: it can also arise in regular circuit families and under local-design assumptions, depending on depth, locality, and observable structure [21], [22], [23].

Given the importance of large qubit numbers for

Received: 2025-12-26; Received in Revised Form: 2026-02-24; Accepted: 2026-02-24.

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Financial Support: The national research foundation of Singapore (NRF-P2024-001).

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DOI: 10.23919/AISE.2026.000005

quantum advantage, mitigating BP is a prerequisite for scalable VQAs. Several approaches have been developed, including restricting to shallow circuits or special architectures where concentration was provably weakened [18], [19], [20], [21], [22], [23], [24]. However, shallow circuits may introduce other landscape difficulties, such as pronounced gaps between local and global optima [25], [26]. In addition, shallow-depth regimes can fall within the reach of classical simulation methods such as tensor networks, which may compromise the prospect of quantum advantage for certain tasks [27], [28]. As a result, practical VQAs design faces a persistent dilemma: increasing depth can improve expressibility and representational power, yet it often exacerbates trainability.

In this work, we resolve this dilemma for a broad and practically relevant setting by providing architectural solutions to BP for deep variational quantum circuits. Specifically, we prove explicit lower bounds on ① the second moment of the loss and ② the second moment of the gradient norm as functions of the circuit depth and width. These bounds quantify that, under our architectural conditions, the relevant second-order statistics do not vanish exponentially with system size. We further relate our constructions to existing BP-mitigation architectures and show that several prior designs can be viewed as shallow special cases of our frameworks. Finally, we provided numerical validations across multiple tasks: compared to hardware-efficient circuits [29] and deep random circuits with comparable parameter counts, quantum controlled-layer (QCL) and quantum residual network (QResNet) consistently exhibit larger initial gradients, faster convergence, and better final objective values.

The BP phenomenon was first formalized for global-design random circuits [17], showing exponentially vanishing gradient variance in N . Later works relaxed global 2-design assumptions to local designs and structured circuits, relating gradient concentration to circuit depth, locality, and observable support [21], [22], [24]. Complementary analyses connect BP to expressibility or entanglement and to proximity

to Haar-randomness [23], [32]. A major limitation of much early BP theory was its reliance on Haar-randomness or unitary-design assumptions that were often misaligned with practical parameterized quantum circuits (PQCs) trained over continuous rotation angles. Recent work has developed more structural frameworks, including Lie-algebraic [30] and symmetry-based [33] approaches, that characterize when and why deep parameterized circuits exhibit (or avoid) BP without requiring 2-design hypotheses. In parallel, noise-induced barren plateaus have been studied extensively [31], [34]. A review of the barren plateau can be found in [18].

Recently, some techniques have been proposed to address the barren plateau problem, including the special initialization strategy [35], [36] and the layerwise training method [37]. We remark that these techniques rely on the assumption of low-depth quantum circuits. Specifically, [35] initialized parameters such that the initial quantum circuit is equivalent to an identity matrix ($D = 0$). [37] trained subsets of parameters for each layer so that a low-depth circuit is optimized during the training.

However, it is increasingly recognized that being “provably BP-free” does not automatically imply practical quantum advantage. Architectural constraints that ensure non-vanishing gradients may also correlate with classical simulability in some regimes. This motivates a broader call to rethink what trainability guarantees should mean for variational quantum computing [38]. Our work contributes to this discussion by providing explicit deep, block-structured circuit families with provable non-vanishing gradients in a parameter-space setting, while allowing arbitrarily deep subcircuits within blocks, thereby separating circuit depth from trainability under local observables.

The rest of this paper is organized as follows. Preliminaries of VQAs are introduced in Section 2. The theoretical results are presented in Section 3. We show numerical results in Section 4, which includes the toy model study (Section 4.1), the quantum variational autoencoder (Section 4.2), and the binary classification (Section 4.3). We summarize our conclusion in Section 5.

Table 1 Representative Theoretical Results on Barren Plateaus and Trainability

Structure	Distribution / Assumption	Representative Gradient Statement
Random circuit [17]	Global 2-design	$\text{Var}[\partial_{\theta_k} f] \approx O(2^{-3N})$
Alternating layered [21]	Local 2-design	$\text{Var}[\partial_{\theta_k} f] \gtrsim O(2^{-2m(D+2)})$
QCNN [24]	Local 2-design	$\text{Var}[\partial_{\theta_k} f] \gtrsim O(1/\text{poly}(N))$
Arbitrary circuit [23]	Circuit distribution	$\text{Var}[\partial_{\theta_k} f] \leq 2^{-O(N)} + O(\epsilon^2)$
Noisy depth- D circuit [31]	Noise channel model	$ \partial_{\theta_k} f \leq O(N^{1/2} q^D)$
Dynamical lie algebra [30]	Approximate 2-design on $G = e^{\mathbf{g}}$	$\text{Var}[\partial_{\theta_k} f] = O(1/\dim(\mathbf{g}))$

Note: N : number of qubits; D : depth; m : locality scale of gates/observables; ϵ : distance to Haar-randomness [23]; $q < 1$: noise parameter; $\mathbf{g}$: dynamical Lie algebra [30] of a quantum circuit.

2 Preliminary

2.1 Notations and the Basic Quantum Computing

We use $[N]$ to denote the set $\{1, 2, \dots, N\}$. The form $\|\cdot\|$ denotes the ℓ_2 norm of the vector. We denote a_j as the j -th component of the vector $\mathbf{a}$. The tensor product operation is denoted as “ $\otimes$ ”. The conjugate transpose of a matrix $\mathbf{A}$ is denoted as $\mathbf{A}^\dagger$. The trace of a matrix $\mathbf{A}$ is denoted as $\text{Tr}[\mathbf{A}]$. We denote $\nabla_\theta f$ as the gradient of the function f with respect to the variable θ . We employ notations $\mathcal{O}$ and Ω to describe the standard complexity and the complexity ignoring minor terms, respectively.

Now we introduce quantum computing knowledge. The single-qubit state could be written as the unit vector $|\phi\rangle = a|\mathbf{0}\rangle + b|\mathbf{1}\rangle$, where $a, b \in \mathbb{C}$ and $\{|\mathbf{0}\rangle = (1, 0)^T, |\mathbf{1}\rangle = (0, 1)^T\}$. The n -qubit space is formed by the tensor product of n single-qubit spaces. Given the state $|\mathbf{x}\rangle$, the corresponding density matrix is defined as $\rho = |\mathbf{x}\rangle\langle\mathbf{x}|$, in which $\langle\mathbf{x}| = (|\mathbf{x}\rangle)^\dagger$. A single-qubit operation on the state behaves like the matrix-vector multiplication and can be referred to as the gate ($\text{---}\square\text{---}$), in the circuit language. Specifically, single-qubit operations are often used as $\mathbf{R}_G(\theta) = e^{-i\theta G}$ with $G \in \{X, Y, Z\}$, where

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Pauli matrices $\{I, X, Y, Z\}$ could be referred to as $\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$. Moreover, the controlled-NOT (CNOT) gate and the controlled-Z (CZ) gate are employed for generating quantum entanglement:

$$\text{CNOT} = \text{---}\bullet\text{---} = |\mathbf{0}\rangle\langle\mathbf{0}| \otimes \sigma_0 + |\mathbf{1}\rangle\langle\mathbf{1}| \otimes \sigma_1 \quad (1)$$

$$\text{CZ} = \text{---}\bullet\text{---} = |\mathbf{0}\rangle\langle\mathbf{0}| \otimes \sigma_0 + |\mathbf{1}\rangle\langle\mathbf{1}| \otimes \sigma_3 \quad (2)$$

We could obtain information from quantum states by measurements, for example, measuring the state $|\phi\rangle = a|\mathbf{0}\rangle + b|\mathbf{1}\rangle$ generates 0 and 1 with probability $p(0) = |a|^2$ and $p(1) = |b|^2$, respectively. This operation is equivalent to calculating the average of the observable $\mathbf{O} = \sigma_3$ under the state $|\phi\rangle$:

$$\langle\phi|\mathbf{O}|\phi\rangle \equiv \text{Tr}[\sigma_3|\phi\rangle\langle\phi|] = |a|^2 - |b|^2 = p(0) - p(1) \quad (3)$$

2.2 Variational Quantum Algorithms

In general, the variational quantum algorithm is a classical-quantum hybrid framework that utilizes trainable parameterized quantum circuits. Typical VQAs include the variational quantum eigensolver [13], [39] and quantum neural networks [6], [40], [41], [42]. The crucial part of VQAs is to optimize the function $f(\theta)$ with the form:

$$f(\theta; \rho_{\text{in}}) = \text{Tr}[\mathbf{O}V(\theta)\rho_{\text{in}}V(\theta)^\dagger] \quad (4)$$

where $\mathbf{O}$ denotes the quantum observable and ρ_{in} denotes the density matrix of the input state. We use the linear combination of tensor products of Pauli

matrices to represent the quantum observable. For convenience, we define the locality of a Pauli product observable as the number of qubits such that the corresponding component in the observable is not the identity. For the quantum machine learning (QML) scenario, ρ_{in} encodes the information of the training data, while the observable $\mathbf{O}$ could be a simple σ_3 for classification tasks [42], [43] or a complex form for quantum kernel methods [44], [45]. For the variational quantum eigensolver (VQE) scenario, such as quantum chemistry and quantum simulation, ρ_{in} is usually initialized as $(|\mathbf{0}\rangle\langle\mathbf{0}|)^{\otimes N}$, while $\mathbf{O}$ encodes the problem Hamiltonian.

Based on whether the formulation of problem Hamiltonians is provided, VQAs could use Hamiltonian informed ansatzes [39], [46] or Hamiltonian agnostic ansatzes [13]. For the latter case, a practical tactic [4], [42], [47], [48], [49], [50] is to encode each parameter as the phase of single-qubit gates $\{e^{-i\theta\sigma_k}, k \in \{1, 2, 3\}\}$ while employing two-qubit gates like {CNOT, CZ} among them to generate quantum entanglement. We remark that the partial derivative could be calculated subsequently by using the parameter shifting rule [51], [52],

$$\frac{\partial f}{\partial \theta_j} = f(\theta_+; \rho_{\text{in}}) - f(\theta_-; \rho_{\text{in}}) \quad (5)$$

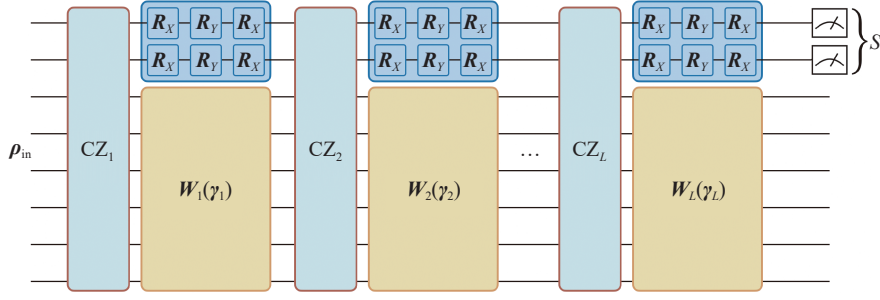
where θ_+ and θ_- are different from θ only at the j th parameter: $\theta_j \rightarrow \theta_j \pm \frac{\pi}{4}$. Thus, the gradient of f could be obtained by estimating quantum observables, which allows optimizations using gradient-based methods [53], [54], [55].

3 Main Results

In this section, we introduce QCL and QResNet architectures and corresponding trainability results. Namely, we prove that the expectation of the square of the gradient for QCL circuits is lower bounded by $\Omega((\frac{1}{8})^{LS})$, where L is the number of controlled layer (CL) blocks in QCL circuits and S is the locality of the quantum observable. For the QResNet circuit, the lower bound scales as $\Omega((\frac{3}{64})^{LS})$, where L is the number of QResNet blocks. Note that the circuit depth in each CL or QResNet block is arbitrary. The bounds are independent of the circuit depth, i.e., we guarantee the trainability of arbitrarily deep QCL and QResNet circuits with limited QCL or QResNet blocks.

3.1 Quantum Controlled-Layer Circuits

Now we discuss the QCL circuit shown in Fig. 1 in detail. The circuit begins with an input state ρ_{in} , followed by L layers of CL blocks. In the ℓ -th CL block, a CZ operator layer (denoted as CZ_ℓ), where CZ gates on arbitrary qubit pairs are allowed, is applied. Then we perform parameterized quantum



Note: Each CL block consists of a CZ operation layer, single-qubit rotations $R_X R_Y R_X$ on the first S qubits, and a parameterized unitary operation $W_\ell(\gamma_\ell)$ on the remaining $N-S$ qubits. Operations CZ_ℓ and $W_\ell(\gamma_\ell)$ could have different structures for different blocks, and the ansatz $W_\ell(\gamma_\ell)$ could be arbitrarily deep.

Fig. 1 The Architecture of the L Blocks of Quantum Controlled-Layer (QCL) Circuits ($S = 2$)

gates on the first S qubits and the remaining $N-S$ qubits separately. For each of the first S qubits, we employ $R_X(\theta_{j,\ell}^{(3)})R_Y(\theta_{j,\ell}^{(2)})R_X(\theta_{j,\ell}^{(1)})$ gates sequence to implement arbitrary unitaries in $\mathcal{U}(2)$ individually, where $j \in \{1, \dots, S\}$ denotes the index of the qubit and $\ell \in \{1, \dots, L\}$ denotes the index of the CL block. We allow arbitrary parameterized unitaries $W_\ell(\gamma_\ell) \in \mathcal{U}(2^{N-S})$ on the remaining $N-S$ qubits. Thus, the ℓ -th CL block is defined by

$$V_\ell(\theta_\ell) = \left(\left(\bigotimes_{j=1}^S W_{j,\ell}(\theta_{j,\ell}) \right) \bigotimes W_\ell(\gamma_\ell) \right) CZ_\ell$$

where $W_{j,\ell}(\theta_{j,\ell}) \equiv R_X(\theta_{j,\ell}^{(3)})R_Y(\theta_{j,\ell}^{(2)})R_X(\theta_{j,\ell}^{(1)})$ and $\theta_\ell = (\theta_{1,\ell}, \dots, \theta_{S,\ell}, \gamma_\ell)$. After all unitary operations, we measure the first S qubits to reveal the information from the quantum circuit. Specifically, the objective function is the expectation of the measurement result with the associated S -local observable $\sigma_i = \sigma_{i_1} \otimes \dots \otimes \sigma_{i_S} \otimes \sigma_0 \otimes \dots \otimes \sigma_0$, where $i_j \neq 0, \forall j \in \{1, \dots, S\}$.

We remark that the controlled-layer architecture shown in Fig. 1 is a general framework. Arbitrary structures could be employed for operations CZ_ℓ and W_ℓ , to generate different distributions for specific tasks. Many existing quantum neural networks can be viewed as special cases of QCL circuits with shallow W_ℓ gates, e.g., the layerwise learning circuit [37] in Fig. 2a, the quantum entanglement circuit [56], [57], [58] in Fig. 2b, and the alternating-layered circuit [21] in Fig. 2c. Specifically, these architectures contain a CZ operation layer (orange color) and

the arbitrary parameterized unitaries $W_\ell(\gamma_\ell) \in \mathcal{U}(2^{N-S})$ on the remaining $N-S$ qubits are taken to be just single-qubit rotations. Besides, the local quantum observable condition is naturally satisfied for variational quantum algorithms associated with quantum machine learning [42], combinatorial optimizations [15], [16], and quantum simulations [59], [60].

We provide main theoretical results of QCL circuit in Theorem 3.1. Specifically, we prove that both the square of the loss $f(\theta)$ and the norm of its gradient are lower bounded by $\Omega((\frac{1}{8})^{LS})$. Both bounds are independent of the qubit number N and the circuit depth D , where the latter can be arbitrarily large when employing complex unitaries W_ℓ in Fig. 1. We provide the proof in Appendices B and C.

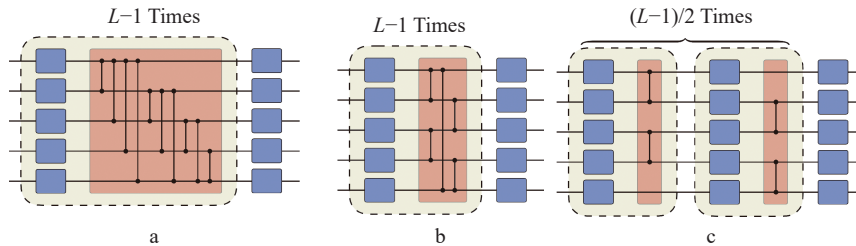
Theorem 3.1 Consider the N -qubit L -block circuit $V(\theta)$ defined in Fig. 1 and the cost function $f(\theta) = \text{Tr}[\sigma_i V(\theta) \rho_{\text{in}} V(\theta)^\dagger]$, where σ_i is a S -local observable. Then the following formulas hold,

$$\mathbb{E}_\theta f^2 \geq \frac{(\text{Tr}[\sigma_{3|i} \rho_{\text{in}}])^2}{8^{LS}} \quad (6)$$

$$\mathbb{E}_\theta \|\nabla_\theta f\|^2 \geq \frac{12(L-1)S}{8^{LS}} (\text{Tr}[\sigma_{3|i} \rho_{\text{in}}])^2 \quad (7)$$

where the index $3|i = (3, \dots, 3, 0, \dots, 0)$ contains S non-zero components. The expectation is taken independently for all parameters in θ with the uniform distribution in $[0, 2\pi]$.

Note that trace terms in Eqs. (6)-(7) are independent of N for many input states, e.g. $\text{Tr}[\sigma_{3|i}(|0\rangle\langle 0|)^{\otimes N}] = 1$.



Note: a. shows the structure of the layerwise learning circuit [37]; b. shows the structure of the quantum entanglement circuit [56], [58]; c. shows the structure of the alternating-layered circuit [37]. The deep blue block denotes the single-qubit rotations and the orange block denotes the CZ operator layer, as defined in Fig. 1.

Fig. 2 Existing Quantum Circuits as Special Cases of QCL Circuits

From the geometric view, the value $\mathbb{E}_\theta \|\nabla_\theta f(\theta)\|^2$ characterizes the global steepness of the function surface in the parameter space. Optimizing the objective function f using gradient-based methods could be hard if the norm of the gradient vanishes to zero. Thus, the lower bound in Eq.(7) provides a theoretical guarantee for the optimization of the objective function, which is the necessary condition for obtaining a good trainability of variational quantum algorithms on related tasks. Different from existing works [17], [21], [24], [35] that deal with shallow quantum circuits, we provide a positive result on the trainability of deep quantum circuits with certain structures.

From the technical view, we provide a new theoretical framework for proving Eqs.(6)-(7). Instead of the unitary 2-design assumptions which do not accord with practical quantum circuits training scenarios, we consider the uniform distribution in the parameter space, in which each parameter in θ varies continuously in $[0, 2\pi]$. Our assumption suits quantum circuits that encode parameters in the phase of single-qubit rotations, and fits the analysis to the loss landscape within the parameter space. Our framework could be extended with further assumptions on current circuit architectures in future works.

3.2 Quantum ResNet

In this subsection, we introduce the QResNet structure and corresponding theoretical results. The QResNet circuit is illustrated in Fig. 3, which contains several QResNet blocks (Fig. 3a). In each QResNet block, all qubits are divided into the source qubits, the target qubits, and the remaining qubits. We begin with $R_X R_Y R_X$ rotations on all source qubits and an arbitrary parameterized unitary $W(\gamma)$ on the remaining qubits. Then we perform CNOT or CZ gates between source and target qubits, i.e. $(i, S+i)$ qubit pairs for $i \in \{1, \dots, S\}$. Next, we perform $R_X R_Y R_X$ rotations on both source and target qubits, followed by two-qubit connections implemented before. Finally, we conduct $R_X R_Y R_X$ rotations on all target qubits. The gate $W(\gamma)$ could have arbitrary structure and depth, so the QResNet circuit could be arbitrarily deep. Fig. 3b shows a QResNet

circuit with 3 QResNet blocks and sequential source and target qubits. Similar to classical residual neural networks [61], QResNet allows efficient information transport through source and target qubits.

We provide main theoretical results of QResNet circuit in Theorem 3.2. Specifically, we prove that both the square of the loss $f(\theta)$ of the QResNet circuit and the norm of its gradient are lower bounded by $\Omega((\frac{3}{64})^{LS})$. Both bounds are independent of the qubit number N and the circuit depth D , where the latter can be arbitrarily large when employing complex unitaries $W(\gamma)$ in Fig. 3b. We provide the proof in Appendices D and E.

Theorem 3.2 Consider the N -qubit circuit $V(\theta)$ defined in Fig. 3 and the function $f(\theta) = \text{Tr}[OV(\theta)\rho_{\text{in}}V(\theta)^\dagger]$, where $O = \sigma_{0^{LS}} \otimes \sigma_i \otimes \sigma_{0^{N-(L+1)S}}$ is a S -local observable. Let $t_j := \text{Tr}[\sigma_j \otimes \sigma_{0^{N-S}} \rho_{\text{in}}]$, then the following formulas hold,

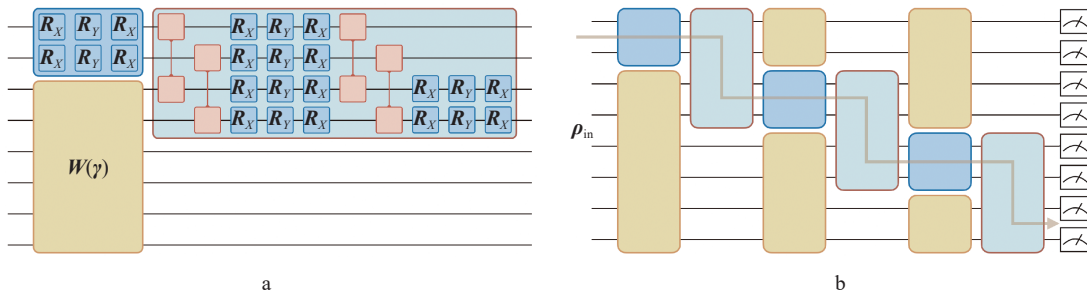
$$\mathbb{E}_\theta f^2 \geq \frac{1}{3} \left(\frac{3}{64}\right)^{LS} \sum_{j \in \{1,2,3\}^{\otimes S}} t_j^2 \quad (8)$$

$$\mathbb{E}_\theta \|\nabla_\theta f\|^2 \geq 4LS \left(\frac{3}{64}\right)^{LS} \sum_{j \in \{1,2,3\}^{\otimes S}} t_j^2 \quad (9)$$

The expectation is taken with the uniform distribution in $[0, 2\pi]$ for all parameters in θ independently.

Remark 1 The lower bounds in Theorems 3.1 and 3.2 exhibit an exponential dependence on the number of blocks L and the locality parameter S . It is important to note that L here counts the macro-level compositional blocks in the proposed QCL and QResNet architectures, whereas the depth within each block can be arbitrarily large. In particular, the bound is independent of the internal depth or the expressivity of sub-circuits W_ℓ : one can increase representation power by deepening W_ℓ without deteriorating the guaranteed gradient scaling.

Remark 2 From a practical perspective, the dependence on S is typically mild in many VQA settings where the readout is local or few-body (e.g., $S \in \{1, 2\}$ for single-qubit/two-qubit measurements



Note: a. shows the structure of a single QResNet circuit ($S = 2$). The circuit consists of $R_X R_Y R_X$ rotations, CNOT or CZ gates between source and target qubits, and an arbitrary parameterized unitary denoted as $W(\gamma)$; b. shows a QResNet circuit with stacked QResNet blocks and sequential source and target qubits ($L = 3$).

Fig. 3 The QResNet Framework

or local correlators). Consequently, the exponential factor should be interpreted as a conservative worst-case guarantee governed by controllable design choices (moderate L and low-locality readout), rather than as an unavoidable manifestation of increasing internal circuit depth. This provides a concrete design principle: keep the number of blocks L moderate to preserve trainability guarantees, and allocate additional expressivity primarily inside the deep sub-circuits W_ℓ .

Limitations While our theoretical analysis establishes non-vanishing lower bounds on relevant second-moment quantities under the proposed architectural conditions, several limitations remain. First, our guarantees characterize trainability in an average-case sense over the parameter space and do not directly address worst-case optimization trajectories or convergence rates under specific optimizers. Second, the analysis focuses on structural properties of the circuit ansatz and does not incorporate noise models or hardware constraints, which may affect practical performance. Finally, although the proposed architectures avoid barren plateaus under the stated assumptions, further investigation is needed to understand trade-offs between trainability, expressivity, and classical simulability in larger-scale settings.

4 Applications

We provided numerical analysis on the trainability of different variational quantum algorithms using the PennyLane Python package [62]. We propose three optimization tasks, i.e., the toy model, the quantum variational autoencoder, and supervised learning for the binary classification. In all tasks, we are able to show that only QCL and QResNet circuits are trainable with deep circuit structures while hardware-efficient circuits [29] and randomly structured circuits are not.

4.1 Toy Model

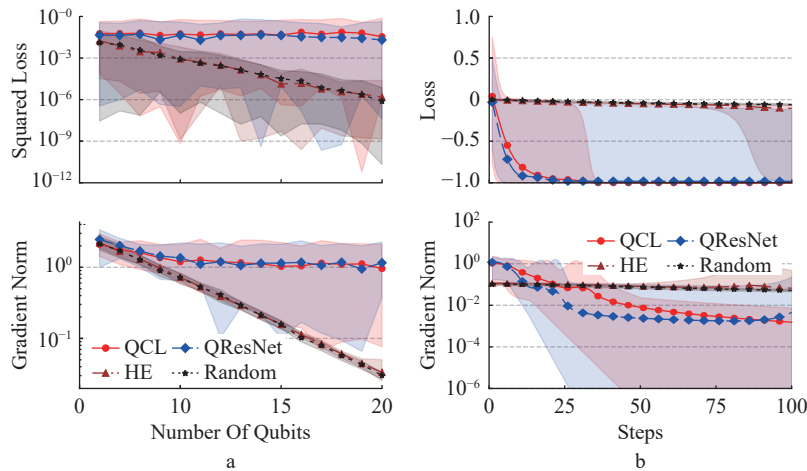
To begin with, we study the optimization property of QCL and QResNet circuits on the toy model,

$$f(\theta) = \text{Tr}[\mathbf{O}V(\theta)(|0\rangle\langle 0|)^{\otimes N}V(\theta)^\dagger] \quad (10)$$

where the observable $\mathbf{O} = \sigma_3 \otimes \sigma_0 \otimes \cdots \otimes \sigma_0$. We compare the performance of four different ansatzes, the QCL circuit, the QResNet circuit, the hardware-efficient (HE) circuit in Fig. 5, and the randomly structured circuit.

For QCL circuits, we set the number of CL blocks $L = 3$ and perform CZ gates on all neighboring qubits $(1,2), (2,3), \dots, (N-1,N), (N,1)$. We have $S = 1$ by considering the observable in Eq.(10). Unitaries on the remaining $N-S$ qubits (the light yellow block in Fig. 1) are chosen to be $L_{HE} = 5$ layers of the HE ansatz in Fig. 5. We remark that the unitary $W(\gamma)$ in the last QCL block would not influence the measurement result of Eq.(10), so the effective depth of the parameterized part of QCL circuits is 30. For QResNet circuits, we set the number of QResNet blocks $L = 1$ and perform CZ gates between source and target qubits. Similar to the QCL circuit, we use $L_{HE} = 10$ layers of the HE ansatz to implement the unitary $W(\gamma)$. Thus, the effective depth of the parameterized part of QCL and QResNet circuits is the same. To make a fair comparison, we conduct the HE circuit with $L_{HE} = 10$ layers, and the number of single-qubit gates and CNOT gates in random circuits is set to be the same as that in HE circuits. Parameters in all circuits are initialized independently from a uniform distribution in $[0, 2\pi]$ if no additional requirements are made.

Numerical results of the toy model are shown in Fig. 4. First, we record the squared loss and the gradient norm of Eq.(10) for QCL circuits, QResNet circuits, HE circuits, and random circuits in Fig. 4a. For HE circuits and random circuits, both the



Note: The left column shows the squared loss and the ℓ_2 -norm of the gradient at the initial stage with increasing qubits $N \in \{6, 7, \dots, 20\}$, respectively. The right column shows the loss value and the ℓ_2 -norm of the gradient during the gradient descent optimization for the case $N = 16$, respectively. Each data point denotes the average of 50 rounds of independent simulations.

Fig. 4 Numerical Results of the Loss Value and the Gradient Information of the Toy Model in Eq.(10)

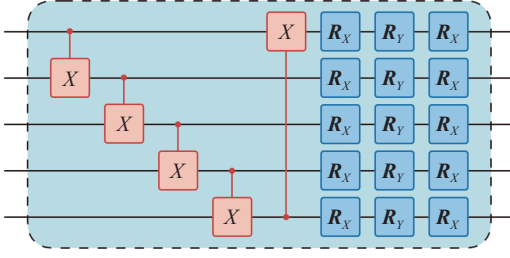


Fig. 5 An Illustration of the Hardware-Efficient Circuit [29]

squared loss and the gradient norm decay exponentially with increasing qubit numbers $N \in \{6, 7, \dots, 20\}$, while for QCL circuits and QResNet circuits, these values remain at the same magnitude. Thus, we have verified Theorems 3.1-3.2. Next, we conduct the gradient descent optimization of Eq.(10) with different circuits. We set the qubit number $N=16$, the learning rate $\eta=0.1$, and the training iteration $T=100$. The result is illustrated in Fig. 4b. Both the QCL circuit and the QResNet circuit converge to the global minimum -1 quickly after around 20 iterations, while the HE circuit and the random circuit have a final loss value that is higher than -0.15 . The initial gradient norm of the QCL circuit and the QResNet circuit is larger than that of the HE circuit and the random circuit by around one order of magnitude, which leads to faster convergence rates.

4.2 Variational Autoencoder

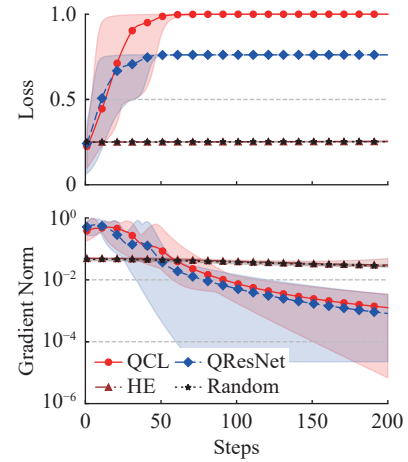
In the second task, we tested the performance of QCL and QResNet circuits on the quantum variational autoencoder (QVA) task [63]. Given a group of m N -qubit input states $\{|\phi_i\rangle\}_{i=1}^m$, the circuit $V(\theta)$, and the number of latent qubits K , the quantum variational autoencoder compresses the original state space $\mathcal{H}(2^N)$ into the latent state space $\mathcal{H}(2^K)$. The corresponding optimization problem is

$$\begin{aligned} & \max_{\theta} f_{\text{QVA}}(\theta) \\ & = \max_{\theta} \frac{1}{m} \sum_{i=1}^m \text{Tr}[\mathbf{O}_K V(\theta) |\phi_i\rangle \langle \phi_i| V(\theta)^\dagger] \end{aligned} \quad (11)$$

where $\mathbf{O}_K = (|0\rangle\langle 0|)^{\otimes(N-K)} \otimes \sigma_0^{\otimes K}$ is an $(N-K)$ -local observable. The optimal value of Eq. (11) is 1, which means the information of the original input states is compressed into the final states perfectly with fewer qubits.

In the experiment, we generate $m=10$ states from the random circuit with the qubit number $N=16$. The number of single-qubit rotations and CNOT gates in the random circuit is 32 and 10, respectively. We set the number of latent qubits $K=14$ and use $S=N-K=2$ in QCL and QResNet circuits. Other gate settings are the same as those in the toy model experiment. We set the number of training iterations as 200 and the learning rate as $\eta=0.1$.

The numerical result of the QVA task is shown in Fig. 6. The QCL and QResNet circuits achieve better performances on the QVA task than the HE circuit and the random circuit. Specifically, the autoencoder using the QCL circuit converges to 1 quickly after around 50 iterations. The QResNet-based autoencoder converges to a lower final value around 0.8. The autoencoder using the HE circuit and the random circuit has the worst final value, which is below 0.3. Besides, the gradient norm of the QCL and the QResNet circuits at initial stage is larger than that of the HE circuits and the random circuit by one order of magnitude. Thus, the optimization of the QCL and QResNet circuits is free from barren plateaus.



Note: The first and second figures show the value of the loss function and the ℓ_2 -norm of the gradient during the gradient ascent optimization. Each line denotes the average of 10 rounds of independent simulations.

Fig. 6 Numerical Results of Quantum Variational Autoencoders with Different Circuits

4.3 Binary Classification

In the third task, we test the performance of different quantum circuits on classifying two kinds of wines based on their features. The wine dataset [64] consists of 138 instances derived from three different cultivars. Each of the instances has 13 features that measure different chemical components. In this task, we choose two kinds of wines and divide the original dataset into the training set and the test set. Both datasets contain 58 samples, and the number of samples from two different classes is the same. Specifically, we denote $A = \{(\mathbf{x}^{(i)}, y^{(i)})\}$ as the set of input data, where $y^{(i)} = \pm 1$ for different labels. Qubit embedding [65] is employed to encode the information of datasets into quantum states, i.e.,

$\rho_{\text{in}}(\mathbf{x}) = \bigotimes_{j=1}^{\dim(\mathbf{x})} R_Y(x_j) |0\rangle\langle 0| R_Y(x_j)^\dagger$, where $\dim(\mathbf{x}) = 13$ is the dimension of $\mathbf{x}$. Let $\hat{y}(\theta; \mathbf{x}) := \text{Tr}[\mathbf{O} V(\theta) \rho_{\text{in}}(\mathbf{x}) V(\theta)^\dagger]$, we optimize the loss defined by

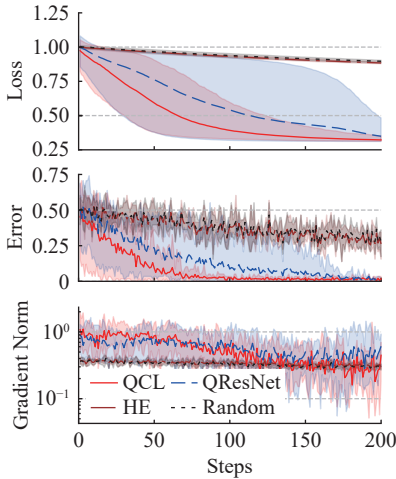
$$f_{\text{QML}}(\theta) = \frac{1}{|A|} \sum_{(x,y) \in A} (\hat{y}(\theta; x) - y)^2 \quad (12)$$

where $\mathbf{O} = \sigma_3 \otimes \sigma_0 \otimes \cdots \otimes \sigma_0$, and $|A|$ is the size of the dataset. We record the classification error

$$e(\theta) = \frac{1}{2|A|} \sum_{(x,y) \in A} [1 - \text{sign}(\hat{y}(\theta; x)y)] \quad (13)$$

on the test dataset.

We show the performance of QCL, QResNet, HE, and random circuits in Fig. 7. The qubit number $N = 13$ suffices to encode all features of the wine dataset. We have $S = 1$ by considering the observable in Eq. (12). We use $L_{\text{HE}} = 10$ for QCL and QResNet circuits with numbers of CL or QResNet blocks as $L = 3$ and $L = 2$, respectively. Other gate settings are the same as those in the toy model experiment. Parameters in different circuits are initialized independently from a uniform distribution in $[0, 2\pi]$. The number of training iterations is 200. The learning rate and the batch size are 0.01 and 8, respectively.



Note: The first figure shows the loss of the training set. The second figure shows the error of the test set. The third figure shows the ℓ_2 -norm of the corresponding gradient during the training. Each line denotes the average of 5 rounds of independent simulations.

Fig. 7 Numerical Results of the Binary Classification on the Wine Dataset [64] Using Stochastic Gradient Descent

The numerical result of the binary classification task using stochastic gradient descent is shown in Fig. 7. For the QCL circuit and the QResNet circuit, both the training loss and the test error decrease significantly and converge to zero compared with those of the HE circuit and the random circuit with the final error around 0.3. The gradient norms of the QCL circuit and the QResNet circuit are around 1.0 at the initial stage and converge during the training. On the other hand, the gradient norm of the HE circuit and the random circuit is smaller than that of the QCL circuit and the QResNet circuit at the initial stage before the latter converges.

In conclusion, deep QCL and QResNet circuits have shown much better trainability for the quantum variational autoencoder and the binary classification when compared with random circuits and HE circuits. These experiments accord with the theorems obtained in the previous section.

5 Conclusion

In this work, we introduced deep QCL and QResNet circuit architectures and provided rigorous lower bounds on the second moments of the loss and gradient norm, demonstrating that these architectures avoid barren plateaus under broad conditions. Importantly, our results show that trainability can be preserved even as the internal depth of each block increases, thereby decoupling expressivity from the typical gradient suppression observed in generic random circuits.

Our numerical experiments on a toy model, a quantum variational autoencoder, and a supervised learning task further corroborate the theoretical findings, showing that QCL and QResNet circuits maintain stable gradient magnitudes and achieve significantly better optimization performance compared with hardware-efficient and randomly structured circuits.

Future research directions include extending the analysis to noisy settings, exploring the relationship between trainability and classical simulability, and investigating adaptive or problem-specific block constructions for practical quantum machine learning applications. For example, recent studies on structured and coupled dynamical systems highlighted that architectural design can critically shape system-level properties such as stability and information propagation [66], [67], [68]. It would be interesting to explore whether such structural design principles can be leveraged to guide the construction of trainable quantum circuit architectures.

APPENDIX A Technical Lemmas

In this section we provide some technical lemmas.

Consider the objective function of the variational quantum algorithm defined as

$$f(\theta) = \langle \mathbf{O} \rangle = \text{Tr}[\mathbf{O}V(\theta)\rho_{\text{in}}V(\theta)^\dagger]$$

where θ encodes all parameters which participate in the circuit as $e^{-i\theta_j\sigma_k}$, $k \in 1, 2, 3$, ρ_{in} denotes the input state and $\mathbf{O}$ is an arbitrary quantum observable.

Lemma A.1 The partial derivative of the function $f(\theta)$ with respect to the parameter θ_j could be calculated by

$$\frac{\partial f}{\partial \theta_j} = \text{Tr}[\mathbf{O}V(\theta_+) \rho_{\text{in}} V(\theta_+)^\dagger] - \text{Tr}[\mathbf{O}V(\theta_-) \rho_{\text{in}} V(\theta_-)^\dagger]$$

where $\theta_+ \equiv \theta + \frac{\pi}{4}e_j$ and $\theta_- \equiv \theta - \frac{\pi}{4}e_j$.

Proof First we assume that the circuit $V(\theta)$ consists of p parameters, and could be written in the sequence:

$$V(\theta) = V_p(\theta_p)V_{p-1}(\theta_{p-1})\cdots V_1(\theta_1)$$

such that each block V_j contains only one parameter.

The parameter shifting rule [51] provides a gradient calculation method for the single parameter case. Consider the observable $\mathcal{O}' = V_{j+1}^\dagger \cdots V_p^\dagger \mathcal{O} V_p \cdots V_{j+1}$ and the input state $\rho'_{\text{in}} = V_{j-1} \cdots V_1 \rho_{\text{in}} V_1^\dagger \cdots V_{j-1}^\dagger$. The gradient of $f_j(\theta_j) = \text{Tr}[\mathcal{O}' U(\theta_j) \rho'_{\text{in}} U(\theta_j)^\dagger]$ could be calculated as

$$\frac{df_j}{d\theta_j} = f_j(\theta_j + \frac{\pi}{4}) - f_j(\theta_j - \frac{\pi}{4})$$

Thus, by inserting the form of $\mathcal{O}'$ and ρ'_{in} , we could obtain

$$\begin{aligned} \frac{\partial f}{\partial \theta_j} &= \frac{df_j}{d\theta_j} \\ &= f_j(\theta_j + \frac{\pi}{4}) - f_j(\theta_j - \frac{\pi}{4}) \\ &= \text{Tr}[\mathcal{O} V(\theta_+) \rho_{\text{in}} V(\theta_+)^\dagger] - \text{Tr}[\mathcal{O} V(\theta_-) \rho_{\text{in}} V(\theta_-)^\dagger] \end{aligned}$$

Lemma A.2 Let θ be a variable with uniform distribution in $[0, 2\pi]$. Let G be an arbitrary hermitian unitary, and define $W = e^{-i\theta G}$. Let $\mathcal{O}$ be an arbitrary quantum observable. Then

$$\begin{aligned} \mathbb{E}_\theta \text{Tr}[\mathcal{O} W \rho_1 W^\dagger] \text{Tr}[\mathcal{O} W \rho_2 W^\dagger] &= \text{Tr}[\mathcal{O}_1 \rho_1] \text{Tr}[\mathcal{O}_1 \rho_2] + \\ &\frac{1}{2} \text{Tr}[\mathcal{O}_2 \rho_1] \text{Tr}[\mathcal{O}_2 \rho_2] + \frac{1}{2} \text{Tr}[i\mathcal{O}_2 G \rho_1] \text{Tr}[i\mathcal{O}_2 G \rho_2] \end{aligned} \quad (14)$$

where ρ_1, ρ_2 are two linear sum of quantum density matrices, and $\mathcal{O}_{1,2} = \frac{1}{2} \mathcal{O} \pm \frac{1}{2} G \mathcal{O} G$, respectively.

Proof By replacing the term

$$W = e^{-i\theta G} = I \cos \theta - iG \sin \theta$$

we have

$$\begin{aligned} \text{Tr}[\mathcal{O} W \rho_1 W^\dagger] &= \text{Tr}[\mathcal{O} (I \cos \theta - iG \sin \theta) \rho_1 (I \cos \theta + iG \sin \theta)] \\ &= \text{Tr}[\mathcal{O}_1 \rho_1] + \cos 2\theta \text{Tr}[\mathcal{O}_2 \rho_1] - \sin 2\theta \text{Tr}[i\mathcal{O}_2 G \rho_1] \end{aligned} \quad (15)$$

where hermitians $\mathcal{O}_1$ and $\mathcal{O}_2$ are commuting and anti-commuting parts of $\mathcal{O}$ with respect to G , respectively, i.e.

$$\begin{aligned} \mathcal{O}_1 + \mathcal{O}_2 &= \mathcal{O} \\ \mathcal{O}_1 G - G \mathcal{O}_1 &= \frac{\mathcal{O} G + G \mathcal{O}}{2} - \frac{G \mathcal{O} + \mathcal{O} G}{2} = 0 \\ \mathcal{O}_2 G + G \mathcal{O}_2 &= \frac{\mathcal{O} G - G \mathcal{O}}{2} + \frac{G \mathcal{O} - \mathcal{O} G}{2} = 0 \end{aligned}$$

We remark that $i\mathcal{O}_2 G$ could serve as a hermitian observable. Similar formulation also holds for

ρ_2 , i.e.

$$\begin{aligned} \text{Tr}[\mathcal{O} W \rho_2 W^\dagger] &= \text{Tr}[\mathcal{O}_1 \rho_2] + \cos 2\theta \text{Tr}[\mathcal{O}_2 \rho_2] - \sin 2\theta \text{Tr}[i\mathcal{O}_2 G \rho_2] \end{aligned} \quad (16)$$

Combining Eqs. (15) - (16), we have

$$\begin{aligned} \mathbb{E}_\theta \text{Tr}[\mathcal{O} W \rho_1 W^\dagger] \text{Tr}[\mathcal{O} W \rho_2 W^\dagger] &= \mathbb{E}_\theta (\text{Tr}[\mathcal{O}_1 \rho_1] + \cos 2\theta \text{Tr}[\mathcal{O}_2 \rho_1] - \sin 2\theta \text{Tr}[i\mathcal{O}_2 G \rho_1]) \cdot \\ &(\text{Tr}[\mathcal{O}_1 \rho_2] + \cos 2\theta \text{Tr}[\mathcal{O}_2 \rho_2] - \sin 2\theta \text{Tr}[i\mathcal{O}_2 G \rho_2]) \\ &= \text{Tr}[\mathcal{O}_1 \rho_1] \text{Tr}[\mathcal{O}_1 \rho_2] + \frac{1}{2} \text{Tr}[\mathcal{O}_2 \rho_1] \text{Tr}[\mathcal{O}_2 \rho_2] + \\ &\frac{1}{2} \text{Tr}[i\mathcal{O}_2 G \rho_1] \text{Tr}[i\mathcal{O}_2 G \rho_2] \end{aligned} \quad (17)$$

where Eq. (17) is obtained by calculating expectation terms.

Lemma A.3 Let θ be a variable with uniform distribution in $[0, 2\pi]$. Let G be an arbitrary hermitian unitary, and denote $W_\pm = e^{-i(\theta \pm \frac{\pi}{4})G}$. Let $\mathcal{O}$ be an arbitrary hermitian observable and ρ be a quantum state. Then

$$\begin{aligned} \mathbb{E}_\theta (\text{Tr}[\mathcal{O} W_+ \rho W_+^\dagger] - \text{Tr}[\mathcal{O} W_- \rho W_-^\dagger])^2 &= 2\text{Tr}[\mathcal{O}_2 \rho]^2 + 2\text{Tr}[i\mathcal{O}_2 G \rho]^2 \end{aligned} \quad (18)$$

where $\mathcal{O}_2 = \frac{1}{2} \mathcal{O} - \frac{1}{2} G \mathcal{O} G$.

Proof First we denote $\theta_\pm = \theta \pm \frac{\pi}{4}$ for convenience. By replacing terms

$$W_\pm = e^{-i\theta_\pm G} = I \cos \theta_\pm - iG \sin \theta_\pm$$

we obtain

$$\begin{aligned} \text{Tr}[\mathcal{O} W_+ \rho W_+^\dagger] - \text{Tr}[\mathcal{O} W_- \rho W_-^\dagger] &= \text{Tr}[\mathcal{O} (I \cos \theta_+ - iG \sin \theta_+) \rho (I \cos \theta_+ + iG \sin \theta_+)] - \\ &\text{Tr}[\mathcal{O} (I \cos \theta_- - iG \sin \theta_-) \rho (I \cos \theta_- + iG \sin \theta_-)] \\ &= \text{Tr}[\mathcal{O}_1 \rho] + \cos 2\theta_+ \text{Tr}[\mathcal{O}_2 \rho] - \sin 2\theta_+ \text{Tr}[i\mathcal{O}_2 G \rho] - \\ &\text{Tr}[\mathcal{O}_1 \rho] - \cos 2\theta_- \text{Tr}[\mathcal{O}_2 \rho] + \sin 2\theta_- \text{Tr}[i\mathcal{O}_2 G \rho] \\ &= -2 \sin 2\theta \text{Tr}[\mathcal{O}_2 \rho] - 2 \cos 2\theta \text{Tr}[i\mathcal{O}_2 G \rho] \end{aligned}$$

where hermitians $\mathcal{O}_{1,2} = \frac{1}{2} \mathcal{O} \pm \frac{1}{2} G \mathcal{O} G$ are commuting and anti-commuting parts of $\mathcal{O}$ with respect to G , respectively, i.e.

$$\begin{aligned} \mathcal{O}_1 + \mathcal{O}_2 &= \mathcal{O} \\ \mathcal{O}_1 G - G \mathcal{O}_1 &= \frac{\mathcal{O} G + G \mathcal{O}}{2} - \frac{G \mathcal{O} + \mathcal{O} G}{2} = 0 \\ \mathcal{O}_2 G + G \mathcal{O}_2 &= \frac{\mathcal{O} G - G \mathcal{O}}{2} + \frac{G \mathcal{O} - \mathcal{O} G}{2} = 0 \end{aligned}$$

We remark that $i\mathcal{O}_2 G$ could serve as a hermitian observable. Thus, we have

$$\begin{aligned} & \mathbb{E}_\theta \left(\text{Tr}[\mathbf{O} \mathbf{W}_+ \rho \mathbf{W}_+^\dagger] - \text{Tr}[\mathbf{O} \mathbf{W}_- \rho \mathbf{W}_-^\dagger] \right)^2 \\ &= 2\text{Tr}[\mathbf{O}_2 \rho]^2 + 2\text{Tr}[\mathbf{O}_2 \mathbf{G} \rho]^2 \end{aligned} \quad (19)$$

where Eq.(19) is obtained by calculating expectation terms.

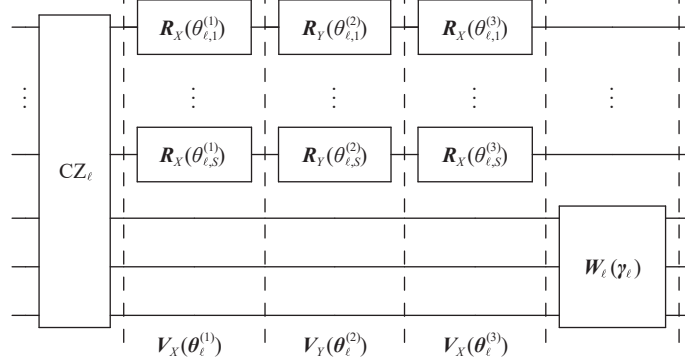


Fig. 8 Decomposition of a Controlled Layer Block into Five Layers

Denote

$$\mathbf{V}_X(\theta_l^{(1)}) \equiv \mathbf{R}_X(\theta_{l,1}^{(1)}) \otimes \cdots \otimes \mathbf{R}_X(\theta_{l,s}^{(1)}) \otimes \sigma_0 \otimes \cdots \otimes \sigma_0 \quad (20)$$

$$\mathbf{V}_Y(\theta_l^{(2)}) \equiv \mathbf{R}_Y(\theta_{l,1}^{(2)}) \otimes \cdots \otimes \mathbf{R}_Y(\theta_{l,s}^{(2)}) \otimes \sigma_0 \otimes \cdots \otimes \sigma_0 \quad (21)$$

$$\mathbf{V}_X(\theta_l^{(3)}) \equiv \mathbf{R}_X(\theta_{l,1}^{(3)}) \otimes \cdots \otimes \mathbf{R}_X(\theta_{l,s}^{(3)}) \otimes \sigma_0 \otimes \cdots \otimes \sigma_0 \quad (22)$$

where $\theta_l^{(i)} = (\theta_{l,1}^{(i)}, \dots, \theta_{l,s}^{(i)})$, $\forall i \in \{1, 2, 3\}$. Then a controlled-layer block in Fig. 8 could be divided into five parts as follows:

$$\mathbf{V}_\ell(\theta_\ell) = (\mathbf{I}^{\otimes S} \otimes \mathbf{W}_\ell(\gamma_\ell)) \mathbf{V}_X(\theta_\ell^{(3)}) \mathbf{V}_Y(\theta_\ell^{(2)}) \mathbf{V}_X(\theta_\ell^{(1)}) \mathbf{CZ}_\ell, \quad (23)$$

where the parameter $\theta_\ell = (\theta_\ell^{(1)}, \theta_\ell^{(2)}, \theta_\ell^{(3)}, \gamma_\ell)$. We abbreviate the term $\mathbf{I}^{\otimes S} \otimes \mathbf{W}_\ell(\gamma_\ell)$ as $\mathbf{W}_\ell(\gamma_\ell)$ in the following for convenience.

In the following proof, we consider $\mathbf{V}_X(\theta_\ell^{(1)})$, $\mathbf{V}_Y(\theta_\ell^{(2)})$, $\mathbf{V}_X(\theta_\ell^{(3)})$, $\mathbf{W}_\ell(\gamma_\ell)$ and $\mathbf{CZ}_\ell$ as separate layers. We denote by ρ_k the state after the k -th layer, $\forall k \in \{0, 1, \dots, 5L\}$,

$$\rho_k := \begin{cases} \left(\prod_{i=\ell}^1 \mathbf{V}_i(\theta_i) \right) \rho_{\text{in}} \left(\prod_{i=1}^\ell \mathbf{V}_i(\theta_i)^\dagger \right) & (k = 5\ell) \\ \mathbf{CZ}_{\ell+1} \rho_{5\ell} \mathbf{CZ}_{\ell+1}^\dagger & (k = 5\ell + 1) \\ \mathbf{V}_X(\theta_{\ell+1}^{(1)}) \rho_{5\ell+1} \mathbf{V}_X(\theta_{\ell+1}^{(1)})^\dagger & (k = 5\ell + 2) \\ \mathbf{V}_Y(\theta_{\ell+1}^{(2)}) \rho_{5\ell+2} \mathbf{V}_Y(\theta_{\ell+1}^{(2)})^\dagger & (k = 5\ell + 3) \\ \mathbf{V}_X(\theta_{\ell+1}^{(3)}) \rho_{5\ell+3} \mathbf{V}_X(\theta_{\ell+1}^{(3)})^\dagger & (k = 5\ell + 4) \end{cases} \quad (24)$$

Now we begin to prove Eq.(6). We denote all parameters in the circuit by $\theta = (\theta_1, \theta_2, \dots, \theta_L)$. Rewrite the formulation of $f(\theta)$ in detail:

$$\mathbb{E}_\theta \left(\text{Tr}[\sigma_i \mathbf{V}(\theta) \rho_{\text{in}} \mathbf{V}(\theta)^\dagger] \right)^2 = \mathbb{E}_\theta \left(\text{Tr}[\sigma_i \rho_{5L}] \right)^2 \quad (25)$$

APPENDIX B Proof of Eq. 6

Proof We first introduce several notations for convenience.

$$= \mathbb{E}_{\theta_1, \dots, \theta_{L-1}, \theta_L^{(1)}, \theta_L^{(2)}, \theta_L^{(3)}, \gamma_L} \text{Tr}[\sigma_i \mathbf{W}_L(\gamma_L) \rho_{5L-1} \mathbf{W}_L(\gamma_L)^\dagger]^2 \quad (26)$$

$$= \mathbb{E}_{\theta_1, \dots, \theta_{L-1}, \theta_L^{(1)}, \theta_L^{(2)}, \theta_L^{(3)}} \text{Tr}[\sigma_i \rho_{5L-1}]^2 \quad (27)$$

$$= \mathbb{E}_{\theta_1, \dots, \theta_{L-1}, \theta_L^{(1)}, \theta_L^{(2)}, \theta_L^{(3)}} \text{Tr}[\sigma_i \mathbf{V}_X(\theta_L^{(3)}) \rho_{5L-2} \mathbf{V}_X(\theta_L^{(3)})^\dagger]^2 \quad (28)$$

Eqs.(25) - (26) follow from the definition of ρ_k in Eq.(24). Eq.(27) holds since

$$\mathbf{W}_L(\gamma_L)^\dagger \sigma_i \mathbf{W}_L(\gamma_L) = \sigma_i \quad (29)$$

Eq.(28) is obtained by using the definition of ρ_k in Eq.(24) again.

Continuing from Eq.(28), we have

$$\begin{aligned} & \mathbb{E}_\theta \left(\text{Tr}[\sigma_i \mathbf{V}(\theta) \rho_{\text{in}} \mathbf{V}(\theta)^\dagger] \right)^2 \\ & \geq \mathbb{E}_{\theta_1, \dots, \theta_{L-1}, \theta_L^{(1)}, \theta_L^{(2)}} \frac{\text{Tr}[\sigma_{3|i;2} \mathbf{V}_Y(\theta_L^{(2)}) \rho_{5L-3} \mathbf{V}_Y(\theta_L^{(2)})^\dagger]^2}{2^S} \end{aligned} \quad (30)$$

$$\geq \mathbb{E}_{\theta_1, \dots, \theta_{L-1}, \theta_L^{(1)}} \frac{\text{Tr}[\sigma_{3|i} \mathbf{V}_X(\theta_L^{(1)}) \rho_{5L-4} \mathbf{V}_X(\theta_L^{(1)})^\dagger]^2}{2^{2S}} \quad (31)$$

$$\geq \frac{1}{2^{3S}} \mathbb{E}_{\theta_1, \dots, \theta_{L-1}} \text{Tr}[\sigma_{3|i} \rho_{5L-4}]^2 \quad (32)$$

$$= \frac{1}{2^{3S}} \mathbb{E}_{\theta_1, \dots, \theta_{L-1}} \text{Tr}[\sigma_{3|i} \rho_{5L-5}]^2 \quad (33)$$

where $3|i;2$ denotes the index by replacing non-zero elements of $\mathbf{i} = (i_1, \dots, i_N)$ with 3 if the original value is 2. Eq.(30) is derived by using Lemma A.2 for the $\mathbf{R}_X$ rotation case for S times. Eqs.(31)-(32) are

derived similarly by using Lemma A.2. Eq.(33) is derived by noticing that for $j, k \in \{0, 3\}$, the term $\sigma_j \otimes \sigma_k$ remains the same after the CZ operation.

Now we move forward from Eq.(33). By applying Eqs.(26)–(33) inductively for $L-1$ times, we obtain

$$\text{Eq. (33)} \geq \frac{1}{2^{3LS}} \text{Tr}[\sigma_{3|i} \rho_{\text{in}}]^2 \quad (34)$$

Thus, we prove Eq.(6).

APPENDIX C Proof of Eq. 7

Proof First, we notice that the norm of the whole gradient is lower bounded by that of particle derivatives summed over a part of parameters, i.e.

$$\mathbb{E}_{\theta} \|\nabla_{\theta} f\|^2 \geq \sum_{\ell=1}^{L-1} \sum_{n=1}^S \sum_{j=1}^3 \mathbb{E}_{\theta} \left(\frac{\partial f}{\partial \theta_{\ell,n}^{(j)}} \right)^2 \quad (35)$$

where $\theta_{\ell,n}^{(j)}$ denotes the parameter of the j th single-qubit gate on the n th qubit in the ℓ th CL block. Thus, we could obtain Eq.(7) if

$$\mathbb{E}_{\theta} \left(\frac{\partial f}{\partial \theta_{\ell,n}^{(j)}} \right)^2 \geq \frac{4(\text{Tr}[\sigma_{3|i} \rho_{\text{in}}])^2}{8^{LS}} \quad (36)$$

holds for all $\ell \in \{1, \dots, L-1\}$, $n \in \{1, \dots, S\}$, and $j \in \{1, 2, 3\}$.

Now we begin to prove Eq.(36). Our main idea is to integrate the square of the partial derivative of f with respect to $\theta = (\theta_1, \dots, \theta_L)$ by using Lemma A.2 and Lemma A.3. For convenience, we follow the definition of $V_X(\cdot)$ in Eqs.(20), (22), $V_Y(\cdot)$ in Eq.(21), CZ_{ℓ} and ρ_k in Eq.(24). In addition, we denote

$$\rho_{k,\ell,n,j,\pm} := \begin{cases} \rho_k(\theta_1, \dots) & (k \leq 5\ell-4) \\ \rho_k(\theta_1, \dots, \theta_{\ell-1}, \theta_{\ell,n,j,\pm}, \theta_{\ell+1}, \dots) & (k > 5\ell-4) \end{cases} \quad (37)$$

where ρ_k follows from the definition in Eq.(24), and $\theta_{\ell,n,j,\pm}$ differs from θ_{ℓ} by plus or minus $\frac{\pi}{4}$ on the component $\theta_{\ell,n}^{(j)}$.

Next, we rewrite the formulation in Eq.(36),

$$\mathbb{E}_{\theta} \left(\frac{\partial}{\partial \theta_{\ell,n}^{(j)}} \text{Tr}[\sigma_i V(\theta) \rho_{\text{in}} V(\theta)^{\dagger}] \right)^2 \quad (38)$$

$$= \mathbb{E}_{\theta_1, \dots, \theta_L} \left(\frac{\partial}{\partial \theta_{\ell,n}^{(j)}} \text{Tr}[\sigma_i \rho_{5L}] \right)^2 \quad (39)$$

$$= \mathbb{E}_{\theta_1, \dots, \theta_L} (\text{Tr}[\sigma_i \rho_{5L,\ell,n,j,+}] - \text{Tr}[\sigma_i \rho_{5L,\ell,n,j,-}])^2 \quad (40)$$

Eq.(38) follows from the formulation of the cost

function f . Eq.(39) follows from Eq.(24). Eq.(40) follows from the parameter-shift rule in Lemma A.1.

We proceed from Eq.(40). Firstly, by taking the expectation with the uniform distribution of parameters in θ_L , we obtain

$$\begin{aligned} & \mathbb{E}_{\theta_L} (\text{Tr}[\sigma_i \rho_{5L,\ell,n,j,+}] - \text{Tr}[\sigma_i \rho_{5L,\ell,n,j,-}])^2 \\ &= \mathbb{E}_{\theta_L^{(1)}, \theta_L^{(2)}, \theta_L^{(3)}, \gamma_L} (\text{Tr}[\sigma_i W_L(\gamma_L) \rho_{5L-1,\ell,n,j,+} W_L(\gamma_L)^{\dagger}] - \text{Tr}[\sigma_i W_L(\gamma_L) \rho_{5L-1,\ell,n,j,-} W_L(\gamma_L)^{\dagger}])^2 \end{aligned} \quad (41)$$

$$= \mathbb{E}_{\theta_L^{(1)}, \theta_L^{(2)}, \theta_L^{(3)}} (\text{Tr}[\sigma_i \rho_{5L-1,\ell,n,j,+}] - \text{Tr}[\sigma_i \rho_{5L-1,\ell,n,j,-}])^2 \quad (42)$$

$$= \mathbb{E}_{\theta_L^{(1)}, \theta_L^{(2)}, \theta_L^{(3)}} (\text{Tr}[\sigma_i V_X(\theta_L^{(3)}) \rho_{5L-2,\ell,n,j,+} V_X(\theta_L^{(3)})^{\dagger}] - \text{Tr}[\sigma_i V_X(\theta_L^{(3)}) \rho_{5L-2,\ell,n,j,-} V_X(\theta_L^{(3)})^{\dagger}])^2 \quad (43)$$

where Eq.(41) follows from the definition of $\rho_{k,\ell,n,j,\pm}$ in Eq.(37). Eq.(42) is obtained by using Eq.(29). Eq.(43) follows from the definition of $\rho_{k,\ell,n,j,\pm}$ in Eq.(37) again.

Then, we could further bound Eq.(43) by

$$\geq \frac{1}{2^S} \mathbb{E}_{\theta_L^{(1)}, \theta_L^{(2)}} (\text{Tr}[\sigma_{3|i;2} \rho_{5L-2,\ell,n,j,+}] - \text{Tr}[\sigma_{3|i;2} \rho_{5L-2,\ell,n,j,-}])^2 \quad (44)$$

$$= \frac{1}{2^S} \mathbb{E}_{\theta_L^{(1)}, \theta_L^{(2)}} (\text{Tr}[\sigma_{3|i;2} V_Y(\theta_L^{(2)}) \rho_{5L-3,\ell,n,j,+} V_Y(\theta_L^{(2)})^{\dagger}] - \text{Tr}[\sigma_{3|i;2} V_Y(\theta_L^{(2)}) \rho_{5L-3,\ell,n,j,-} V_Y(\theta_L^{(2)})^{\dagger}])^2 \quad (45)$$

$$\geq \frac{1}{2^{2S}} \mathbb{E}_{\theta_L^{(1)}} (\text{Tr}[\sigma_{3|i} \rho_{5L-3,\ell,n,j,+}] - \text{Tr}[\sigma_{3|i} \rho_{5L-3,\ell,n,j,-}])^2 \quad (46)$$

$$= \frac{1}{2^{2S}} \mathbb{E}_{\theta_L^{(1)}} (\text{Tr}[\sigma_{3|i} V_X(\theta_L^{(1)}) \rho_{5L-4,\ell,n,j,+} V_X(\theta_L^{(1)})^{\dagger}] - \text{Tr}[\sigma_{3|i} V_X(\theta_L^{(1)}) \rho_{5L-4,\ell,n,j,-} V_X(\theta_L^{(1)})^{\dagger}])^2 \quad (47)$$

$$\geq \frac{1}{2^{3S}} (\text{Tr}[\sigma_{3|i} \rho_{5L-4,\ell,n,j,+}] - \text{Tr}[\sigma_{3|i} \rho_{5L-4,\ell,n,j,-}])^2 \quad (48)$$

$$= \frac{1}{2^{3S}} (\text{Tr}[\sigma_{3|i} \rho_{5L-5,\ell,n,j,+}] - \text{Tr}[\sigma_{3|i} \rho_{5L-5,\ell,n,j,-}])^2 \quad (49)$$

where $3|i;2$ denotes the index by replacing non-zero elements of $i = (i_1, \dots, i_N)$ with 3 if the original value is 2. Eq.(44) is obtained by using Lemma A.2 for the R_X case. Eq.(45) follows from the definition of

$\rho_{k,\ell,n,j,\pm}$ in Eq.(37). Eq.(46) is obtained by using Lemma A.2 for the R_Y case. Eq.(47) follows from the definition of $\rho_{k,\ell,n,j,\pm}$ in Eq.(37). Eq.(48) is obtained by using Lemma A.2 for the R_X case. Eq.(49) is derived by noticing that for $j, k \in \{0, 3\}$, the term $\sigma_j \otimes \sigma_k$ remains the same after the CZ operation.

Next, we repeat the derivation in Eqs.(41)–(49) inductively for parameters $(\theta_{L-2}, \dots, \theta_{\ell+1})$, which yields

$$\begin{aligned} & \mathbb{E}_{\theta} \left(\frac{\partial f}{\partial \theta_{\ell,n}^{(j)}} \right)^2 \\ & \geq \frac{1}{2^{3(L-\ell)S}} \mathbb{E}_{\theta_1, \dots, \theta_{\ell}} \left(\text{Tr}[\sigma_{3|i} \rho_{5\ell-1, \ell, n, j, +}] - \text{Tr}[\sigma_{3|i} \rho_{5\ell-1, \ell, n, j, -}] \right)^2 \end{aligned} \quad (50)$$

$$\begin{aligned} & = \frac{1}{8^{(L-\ell)S}} \mathbb{E}_{\theta_1, \dots, \theta_{\ell}} \left(\text{Tr}[\sigma_{3|i} W_{\ell}(\gamma_{\ell}) \rho_{5\ell-1, \ell, n, j, +} W_{\ell}(\gamma_{\ell})^{\dagger}] - \right. \\ & \quad \left. \text{Tr}[\sigma_{3|i} W_{\ell}(\gamma_{\ell}) \rho_{5\ell-1, \ell, n, j, -} W_{\ell}(\gamma_{\ell})^{\dagger}] \right)^2 \end{aligned} \quad (51)$$

$$\begin{aligned} & = \frac{1}{8^{(L-\ell)S}} \mathbb{E}_{\theta_1, \dots, \theta_{\ell-1}, \theta_{\ell}^{(1)}, \theta_{\ell}^{(2)}, \theta_{\ell}^{(3)}} \left(\text{Tr}[\sigma_{3|i} \rho_{5\ell-1, \ell, n, j, +}] - \right. \\ & \quad \left. \text{Tr}[\sigma_{3|i} \rho_{5\ell-1, \ell, n, j, -}] \right)^2 \end{aligned} \quad (52)$$

Eq.(51) is derived by using the definition of $\rho_{k,\ell,n,j,\pm}$ in Eq.(37). Eq.(52) is derived by using Eq.(29).

Now we proceed to integrate parameter layers $(\theta_{\ell}^{(1)}, \theta_{\ell}^{(2)}, \theta_{\ell}^{(3)})$ and $(\theta_1, \dots, \theta_{\ell-1})$. Remark that the parameter $\theta_{\ell,n}^{(j)}$ is applied on the n th qubit with the observable σ_3 in Eq.(52). Since σ_3 anticommutes with σ_1 and σ_2 , Hamiltonians of parameterized single-qubit gates in the circuit, we could apply Lemma A.2 and Lemma A.3 to simplify the formulation. Thus, Eq.(52) could be further bounded

$$\geq \frac{1}{8^{(L-\ell+1)S}} \mathbb{E}_{\theta_1} \dots \mathbb{E}_{\theta_{\ell-1}} 4 \text{Tr}[\sigma_{3|i} \rho_{5\ell-5}]^2 \quad (53)$$

$$\geq \frac{4}{8^{LS}} \text{Tr}[\sigma_{3|i} \rho_{\text{in}}]^2 \quad (54)$$

Eq.(53) is obtained by noticing the similar collapsed formulation of Lemmas A.2 and A.3 when the gate Hamiltonian anticommutes with the

observable. Eq.(54) follows similar to Eq.(34). Thus, we prove Eq.(36).

APPENDIX D Proof of Eq.8

We remark that Eq.8 considers the circuit containing L QResNet blocks, which could be derived by using L times of the result considering the single QResNet block. Here we provide the lower bound of the single QResNet circuit in Theorem D.1 and the corresponding proof.

Theorem D.1 Consider the N -qubit circuit $V(\theta)$ defined in Fig.9 and the function $f(\theta) = \text{Tr}[\mathbf{O}V(\theta)\rho_{\text{in}}V(\theta)^{\dagger}]$, where $\mathbf{O} = \sigma_{0^S} \otimes \sigma_i \otimes \sigma_{0^{N-S}}$ is a S -local observable. Then the following formula holds,

$$\mathbb{E}_{\theta} f(\theta)^2 \geq \frac{1}{64^S} \sum_{j \in \{1, 2, 3\}^{8S}} (\text{Tr}[\sigma_j \otimes \sigma_{0^{N-S}} \rho_{\text{in}}])^2 \quad (55)$$

The expectation is taken with the uniform distribution in $[0, 2\pi]$ for all parameters in θ independently.

Proof First, we note that the result of the general $S = k$ case can be derived from the $S = 1$ case inductively for k times, since the circuit in Fig.9 with $S = k$ contains k QResNet structures. Next, we prove Eq.(55) for the $S = 1$ case. For convenience, we split the QResNet into six pieces and define ρ_k as the density matrix after each layer,

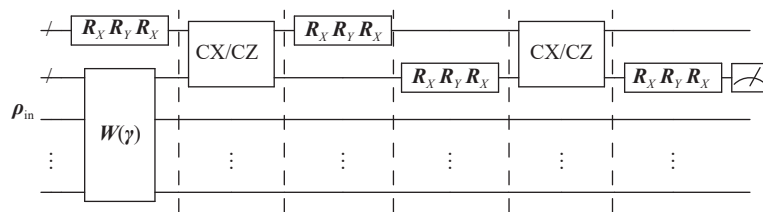
$$\begin{aligned} \rho_1 & = (R_X(\theta_3)R_Y(\theta_2)R_X(\theta_1) \otimes W(\gamma)) \rho_{\text{in}} \cdot \\ & \quad (R_X(\theta_1)^{\dagger} R_Y(\theta_2)^{\dagger} R_X(\theta_3)^{\dagger} \otimes W(\gamma)^{\dagger}) \end{aligned} \quad (56)$$

$$\rho_2 = (G_1 \otimes \sigma_{0^{N-2}}) \rho_1 (G_1^{\dagger} \otimes \sigma_{0^{N-2}}) \quad (57)$$

$$\begin{aligned} \rho_3 & = (R_X(\theta_6)R_Y(\theta_5)R_X(\theta_4) \otimes \sigma_{0^{N-1}}) \rho_2 \cdot \\ & \quad (R_X(\theta_4)^{\dagger} R_Y(\theta_5)^{\dagger} R_X(\theta_6)^{\dagger} \otimes \sigma_{0^{N-1}}) \end{aligned} \quad (58)$$

$$\begin{aligned} \rho_4 & = (\sigma_0 \otimes R_X(\theta_9)R_Y(\theta_8)R_X(\theta_7) \otimes \sigma_{0^{N-2}}) \rho_3 \cdot \\ & \quad (\sigma_0 \otimes R_X(\theta_7)^{\dagger} R_Y(\theta_8)^{\dagger} R_X(\theta_9)^{\dagger} \otimes \sigma_{0^{N-2}}) \end{aligned} \quad (59)$$

$$\rho_5 = (G_2 \otimes \sigma_{0^{N-2}}) \rho_4 (G_2^{\dagger} \otimes \sigma_{0^{N-2}}) \quad (60)$$



Note: The first and the second line denote the ensemble of S wires, where CX or CZ gates are employed on $(1, S+1), \dots, (S, 2S)$ qubits pairs, respectively. The parameterized unitary $W(\gamma)$ acts on wires from the $S+1$ -th to the N -th qubit. The observable is the tensor product of Pauli matrices that acts non-trivially (non-identity) on wires from the $S+1$ -th to the $2S$ -th qubit.

Fig. 9 The Illustration of the N -qubit Quantum ResNet (QResNet) Structure

$$\rho_6 = (\sigma_0 \otimes R_X(\theta_{12})R_Y(\theta_{11})R_X(\theta_{10}) \otimes \sigma_{0^{N-2}}) \rho_5 \cdot$$

$$(\sigma_0 \otimes R_X(\theta_{10})^\dagger R_Y(\theta_{11})^\dagger R_X(\theta_{12})^\dagger \otimes \sigma_{0^{N-2}})$$
(61)

where $G_1, G_2 \in \{|0\rangle\langle 0| \otimes \sigma_0 + |1\rangle\langle 1| \otimes \sigma_1, \sigma_0 \otimes |0\rangle\langle 0| + \sigma_1 \otimes |1\rangle\langle 1|, |0\rangle\langle 0| \otimes \sigma_0 + |1\rangle\langle 1| \otimes \sigma_3\}$ are CX or CZ gates. The parameter $\theta = (\gamma, \theta_1, \dots, \theta_{12})$. By using the definition (61), we could write the left side of Eq.(55) as follows,

$$\mathbb{E}_\theta \text{Tr}[\mathcal{O}\rho_6]^2 = \mathbb{E}_{\gamma, \theta_1, \dots, \theta_{12}} \text{Tr}[\sigma_0 \otimes \sigma_i \otimes \sigma_{0^{N-2}} \rho_6]^2$$

$$\geq \frac{1}{4} \sum_{j \in \{1,2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_0 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_5]^2$$
(63)

where $\sigma_i \in \{\sigma_1, \sigma_2, \sigma_3\}$ and the observable follows the definition in the theorem with $S = 1$. Eq.(63) is obtained by using Lemma A.2 for cases with different σ_i .

Next, we discuss cases with different G_2 . When $G_2 = |0\rangle\langle 0| \otimes \sigma_0 + |1\rangle\langle 1| \otimes \sigma_1$, we have

$$\text{Eq. (63)} \geq \frac{1}{4} \sum_{j \in \{2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_0 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_5]^2$$

$$= \frac{1}{4} \sum_{j \in \{2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_3 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_4]^2$$
(65)

$$\geq \frac{1}{8} \sum_{j \in \{1,2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_3 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_3]^2$$
(66)

$$\geq \frac{1}{32} \sum_{j,k \in \{1,2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_3} \text{Tr}[\sigma_k \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_2]^2$$
(67)

where Eq.(65) is obtained by using the definition (60) and Table 2. Eqs.(66) and (67) are derived by using Lemma A.2.

Table 2 Outputs of CNOT and CZ Gate

$\sigma_j \otimes \sigma_k$	CNOT($\sigma_j \otimes \sigma_k$)CNOT	CZ($\sigma_j \otimes \sigma_k$)CZ
$\sigma_0 \otimes \sigma_0$	$\sigma_0 \otimes \sigma_0$	$\sigma_0 \otimes \sigma_0$
$\sigma_0 \otimes \sigma_1$	$\sigma_0 \otimes \sigma_1$	$\sigma_3 \otimes \sigma_1$
$\sigma_0 \otimes \sigma_2$	$\sigma_3 \otimes \sigma_2$	$\sigma_3 \otimes \sigma_2$
$\sigma_0 \otimes \sigma_3$	$\sigma_3 \otimes \sigma_3$	$\sigma_0 \otimes \sigma_3$
$\sigma_1 \otimes \sigma_0$	$\sigma_1 \otimes \sigma_1$	$\sigma_1 \otimes \sigma_3$
$\sigma_1 \otimes \sigma_1$	$\sigma_1 \otimes \sigma_0$	$\sigma_2 \otimes \sigma_2$
$\sigma_1 \otimes \sigma_2$	$\sigma_2 \otimes \sigma_3$	$-\sigma_2 \otimes \sigma_1$
$\sigma_1 \otimes \sigma_3$	$-\sigma_2 \otimes \sigma_2$	$\sigma_1 \otimes \sigma_0$
$\sigma_2 \otimes \sigma_0$	$\sigma_2 \otimes \sigma_1$	$\sigma_2 \otimes \sigma_3$
$\sigma_2 \otimes \sigma_1$	$\sigma_2 \otimes \sigma_0$	$-\sigma_1 \otimes \sigma_2$
$\sigma_2 \otimes \sigma_2$	$-\sigma_1 \otimes \sigma_3$	$\sigma_1 \otimes \sigma_1$
$\sigma_2 \otimes \sigma_3$	$\sigma_1 \otimes \sigma_2$	$\sigma_2 \otimes \sigma_0$
$\sigma_3 \otimes \sigma_0$	$\sigma_3 \otimes \sigma_0$	$\sigma_3 \otimes \sigma_0$
$\sigma_3 \otimes \sigma_1$	$\sigma_3 \otimes \sigma_1$	$\sigma_0 \otimes \sigma_1$
$\sigma_3 \otimes \sigma_2$	$\sigma_0 \otimes \sigma_2$	$\sigma_0 \otimes \sigma_2$
$\sigma_3 \otimes \sigma_3$	$\sigma_0 \otimes \sigma_3$	$\sigma_3 \otimes \sigma_3$

For other choice of G_2 , we have the same result. When $G_2 = \sigma_0 \otimes |0\rangle\langle 0| + \sigma_1 \otimes |1\rangle\langle 1|$, we have

$$\text{Eq. (63)} \geq \frac{1}{4} \sum_{j \in \{1,2\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_0 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_5]^2$$
(68)

$$= \frac{1}{4} \sum_{j \in \{1,2\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_1 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_4]^2$$
(69)

$$\geq \frac{1}{8} \sum_{j \in \{1,2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_1 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_3]^2$$
(70)

$$\geq \frac{1}{32} \sum_{j,k \in \{1,2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_3} \text{Tr}[\sigma_k \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_2]^2$$
(71)

When $G_2 = |0\rangle\langle 0| \otimes \sigma_0 + |1\rangle\langle 1| \otimes \sigma_3$, we have

$$\text{Eq. (63)} \geq \frac{1}{4} \sum_{j \in \{1,2\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_0 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_5]^2$$
(72)

$$= \frac{1}{4} \sum_{j \in \{1,2\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_3 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_4]^2$$
(73)

$$\geq \frac{1}{8} \sum_{j \in \{1,2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_6} \text{Tr}[\sigma_3 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_3]^2$$
(74)

$$\geq \frac{1}{32} \sum_{j,k \in \{1,2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_3} \text{Tr}[\sigma_k \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_2]^2$$
(75)

Eqs.(68)-(75) are derived similarly to Eqs.(64)-(67). By using Eqs.(62)-(75), we have

$$\mathbb{E}_\theta \text{Tr}[\mathcal{O}\rho_6]^2 \geq \frac{1}{32} \sum_{j,k \in \{1,2,3\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_3} \text{Tr}[\sigma_k \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_2]^2$$
(76)

Next, we discuss on cases with different G_1 . Since j, k in Eq.(76) are symmetrical, we only need to consider one formulation of G_1 for the CX case. When $G_1 = |0\rangle\langle 0| \otimes \sigma_0 + |1\rangle\langle 1| \otimes \sigma_1$, we have

$$\text{Eq. (76)} \geq \frac{1}{32} \sum_{j \in \{1,2\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_3} \text{Tr}[\sigma_j \otimes \sigma_1 \otimes \sigma_{0^{N-2}} \rho_2]^2$$
(77)

$$= \frac{1}{32} \sum_{j \in \{1,2\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_3} \text{Tr}[\sigma_j \otimes \sigma_0 \otimes \sigma_{0^{N-2}} \rho_1]^2$$
(78)

$$\geq \frac{1}{64} \sum_{j \in \{1,2,3\}} \text{Tr}[\sigma_j \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2. \quad (79)$$

Eq.(78) is derived by using Table 2. Eq.(79) is obtained by using the definition (56) and Lemma A.2. When $G_1 = |0\rangle\langle 0| \otimes \sigma_0 + |1\rangle\langle 1| \otimes \sigma_3$, we have

$$\text{Eq. (76)} \geq \frac{1}{32} \sum_{j \in \{1,2\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_3} \text{Tr}[\sigma_j \otimes \sigma_3 \otimes \sigma_{0^{N-2}} \rho_2]^2 \quad (80)$$

$$= \frac{1}{32} \sum_{j \in \{1,2\}} \mathbb{E}_{\gamma, \theta_1, \dots, \theta_3} \text{Tr}[\sigma_j \otimes \sigma_0 \otimes \sigma_{0^{N-2}} \rho_1]^2 \quad (81)$$

$$\geq \frac{1}{64} \sum_{j \in \{1,2,3\}} \text{Tr}[\sigma_j \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 \quad (82)$$

Eq.(81) is derived by using Table 2. Eq.(82) is obtained by using the definition(56) and Lemma A.2. Now we have finished the discussion regarding different G_1 and have proved Theorem D.1.

APPENDIX E Proof of Eq.9

Similar to the situation of Eq.8, Eq.9 could be derived by using $L-1$ times of Theorem D.1 and

$$\mathbb{E}_{\theta} \left(\frac{\partial f(\theta)}{\partial \theta_k} \right)^2 \geq \begin{cases} \frac{5}{64} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{5}{64} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 & (k=1) \\ \frac{3}{64} \text{Tr}[\sigma_1 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{3}{64} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{3}{64} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 & (k=2) \\ \frac{1}{32} \text{Tr}[\sigma_1 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{3}{64} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{3}{64} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 & (k=3) \\ \frac{1}{32} \text{Tr}[\sigma_1 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{1}{32} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{1}{32} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 & (k \in \{5,8\}) \end{cases} \quad (85)$$

Next, we prove Eq.(85). By using the definition (61), we could write the left side of Eq.(85) as follows,

$$\mathbb{E}_{\theta} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\mathbf{O} \rho_6] \right)^2 = \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_{12})} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_0 \otimes \sigma_i \otimes \sigma_{0^{N-2}} \rho_6] \right)^2 \quad (86)$$

$$\geq \frac{1}{4} \sum_{j \in \{1,2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_3)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_0 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_5] \right)^2 \quad (87)$$

where $\sigma_i \in \{\sigma_1, \sigma_2, \sigma_3\}$ and the observable follows the definition in the theorem with $S=1$. Eq.(87) is obtained by using Lemma A.2 for cases with different σ_i . We remark that the derivation of Eq.(87) is very similar to that of Eq.(63). The reason is that the partial derivative of the trace term in Eq.(86) equals to the linear sum of trace terms by using the parameter-shift rule, which is covered by

one time of Theorem E.1, where the latter provides the lower bound of the gradient norm of the single QResNet circuit. Here we provide Theorem E.1 and its proof.

Theorem E.1. Consider the N -qubit circuit $V(\theta)$ defined in Fig.9 and the function $f(\theta) = \text{Tr}[\mathbf{O} V(\theta) \rho_{\text{in}} V(\theta)^\dagger]$, where $\mathbf{O} = \sigma_{0^S} \otimes \sigma_i \otimes \sigma_{0^{N-2S}}$ is a S -local observable. Then the following formula holds,

$$\mathbb{E}_{\theta} \|\nabla_{\theta} f(\theta)\|^2 \geq 4S \left(\frac{3}{64} \right)^S \sum_{j \in \{1,2,3\}^{aS}} \text{Tr}[\sigma_j \otimes \sigma_{0^{N-S}} \rho_{\text{in}}]^2 \quad (83)$$

The expectation is taken with the uniform distribution in $[0, 2\pi]$ for all parameters in θ independently.

Proof Similar to the proof of Theorem E.1, the result of the general $S=k$ case can be derived from the $S=1$ case inductively for k times. Next, we prove Eq.(83) for the $S=1$ case. We follow definitions (56)-(61) to describe intermediate quantum states. Notice that the norm of the whole gradient is lower bounded by that of partial derivatives summed over a part of parameters, i.e.

$$\mathbb{E}_{\theta} \|\nabla_{\theta} f(\theta)\|^2 \geq \sum_{k \in \{1,2,3,5,8\}} \mathbb{E}_{\theta} \left(\frac{\partial f(\theta)}{\partial \theta_k} \right)^2 \quad (84)$$

Thus, we could obtain Eq.(83) for the $S=1$ case if

the condition of Lemma A.2. We could utilize this similarity along with results in Section D to provide a convenient proof. For example, for cases $k \in \{1,2,3\}$, Eq.(87) could be lower bounded by

$$\frac{1}{32} \sum_{j \in \{1,2\}} \mathbb{E}_{(\gamma, \theta_1, \theta_2, \theta_3)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_j \otimes \sigma_{0^{N-1}} \rho_1] \right)^2 \quad (88)$$

by using Eqs.(76)-(78) and Eqs.(80)-(81). For the case $k=3$,

$$\text{Formula(88)} = \frac{1}{32} \mathbb{E}_{(\gamma, \theta_1, \theta_2, \theta_3)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_1] \right)^2 \quad (89)$$

$$= \frac{1}{16} \mathbb{E}_{(\gamma, \theta_1, \theta_2)} \text{Tr}[\mathbf{R}_X(\theta_1)^\dagger \mathbf{R}_Y(\theta_2)^\dagger \sigma_2 \mathbf{R}_Y(\theta_2) \mathbf{R}_X(\theta_1) \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{1}{16} \mathbb{E}_{(\gamma, \theta_1, \theta_2)} \text{Tr}[\mathbf{R}_X(\theta_1)^\dagger \mathbf{R}_Y(\theta_2)^\dagger \sigma_3 \mathbf{R}_Y(\theta_2) \mathbf{R}_X(\theta_1) \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 \quad (90)$$

$$\begin{aligned}
&= \frac{1}{16} \mathbb{E}_{(\gamma, \theta_1)} \text{Tr}[\mathbf{R}_X(\theta_1)^\dagger \sigma_2 \mathbf{R}_X(\theta_1) \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \\
&\quad \frac{1}{32} \mathbb{E}_{(\gamma, \theta_1)} \text{Tr}[\mathbf{R}_X(\theta_1)^\dagger \sigma_3 \mathbf{R}_X(\theta_1) \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \\
&\quad \frac{1}{32} \mathbb{E}_{(\gamma, \theta_1)} \text{Tr}[\mathbf{R}_X(\theta_1)^\dagger \sigma_1 \mathbf{R}_X(\theta_1) \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2
\end{aligned} \quad (91)$$

$$\begin{aligned}
&= \frac{1}{32} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{1}{32} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \\
&\quad \frac{1}{64} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \\
&\quad \frac{1}{64} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{1}{32} \text{Tr}[\sigma_1 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2
\end{aligned} \quad (92)$$

$$\text{Formula(88)} = \begin{cases} \frac{3}{32} \text{Tr}[\sigma_1 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{3}{64} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{3}{64} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 & (k=2) \\ \frac{5}{64} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{5}{64} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 & (k=1) \end{cases} \quad (94)$$

Next, we consider the case $k=8$ and proceed from Eq.(87). Similar to Section D, we discuss on

$$\begin{aligned}
&= \frac{1}{32} \text{Tr}[\sigma_1 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \frac{3}{64} \text{Tr}[\sigma_2 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 + \\
&\quad \frac{3}{64} \text{Tr}[\sigma_3 \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2
\end{aligned} \quad (93)$$

Eq.(89) is derived since σ_1 commutes with the gate $\mathbf{R}_X(\theta_3)$, which induces a zero gradient. Eq.(90) is obtained by using Lemma A.3. Eqs.(91) and (92) are obtained using Lemma A.2. We conduct the same procedure for $k \in \{1, 2\}$ and provide the result below.

different $\mathbf{G}_2$. When $\mathbf{G}_2 = |\mathbf{0}\rangle\langle\mathbf{0}| \otimes \sigma_0 + |\mathbf{1}\rangle\langle\mathbf{1}| \otimes \sigma_1$, we have

$$\text{Eq. (87)} \geq \frac{1}{4} \sum_{j \in \{2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_6)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_0 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_5] \right)^2 \quad (95)$$

$$= \frac{1}{4} \sum_{j \in \{2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_6)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_3 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_4] \right)^2 \quad (96)$$

$$\begin{aligned}
&= \frac{1}{4} \sum_{j \in \{2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_6)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_3 \otimes \mathbf{R}_X(\theta_7)^\dagger \mathbf{R}_Y(\theta_8)^\dagger \mathbf{R}_X(\theta_9)^\dagger \sigma_j \mathbf{R}_X(\theta_9) \mathbf{R}_Y(\theta_8) \right. \\
&\quad \left. \mathbf{R}_X(\theta_7) \otimes \sigma_{0^{N-2}} \rho_3] \right)^2
\end{aligned} \quad (97)$$

$$= \frac{1}{4} \sum_{j \in \{2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_6)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_3 \otimes \mathbf{R}_X(\theta_7)^\dagger \mathbf{R}_Y(\theta_8)^\dagger \sigma_j \mathbf{R}_Y(\theta_8) \mathbf{R}_X(\theta_7) \otimes \sigma_{0^{N-2}} \rho_3] \right)^2 \quad (98)$$

$$= \frac{1}{2} \sum_{j \in \{1,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_7)} \text{Tr}[\sigma_3 \otimes \mathbf{R}_X(\theta_7)^\dagger \sigma_j \mathbf{R}_X(\theta_7) \otimes \sigma_{0^{N-2}} \rho_3]^2 \quad (99)$$

$$= \frac{1}{2} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_6)} \text{Tr}[\sigma_3 \otimes \sigma_1 \otimes \sigma_{0^{N-2}} \rho_3]^2 + \frac{1}{4} \sum_{j \in \{2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_6)} \text{Tr}[\sigma_3 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_3]^2 \quad (100)$$

$$\geq \frac{1}{32} \sum_{j \in \{1,2,3\}} \text{Tr}[\sigma_j \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 \quad (101)$$

Eq.(96) is derived by using the definition (60) and Table 2. Eq.(97) is derived by using the definition (59). Eq.(98) is obtained by integrating over θ_9 via Lemma A.2. Eq.(99) is obtained by integrating θ_8 via Lemma A.3. Eq.(100) is obtained by using

Lemma A.2. We obtain Eq.(101) by using Eqs.(66)-(67) and Eqs.(77)-(82). We remark that Eq.(101) could be obtained for $\mathbf{G}_2 \in \{\sigma_0 \otimes |\mathbf{0}\rangle\langle\mathbf{0}| + \sigma_1 \otimes |\mathbf{1}\rangle\langle\mathbf{1}|, |\mathbf{0}\rangle\langle\mathbf{0}| \otimes \sigma_0 + |\mathbf{1}\rangle\langle\mathbf{1}| \otimes \sigma_3\}$ as well by a derivation similar to Eqs.(95)-(101).

Finally, we consider the case $k = 5$ and proceed from Eq.(87). When $\mathbf{G}_2 = |\mathbf{0}\rangle\langle\mathbf{0}| \otimes \sigma_0 + |\mathbf{1}\rangle\langle\mathbf{1}| \otimes \sigma_1$,

we have

$$\text{Eq. (87)} \geq \frac{1}{8} \sum_{j \in \{1,2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_6)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[\sigma_3 \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_3] \right)^2 \quad (102)$$

$$= \frac{1}{8} \sum_{j \in \{1,2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_6)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[R_X(\theta_4)^\dagger R_Y(\theta_5)^\dagger R_X(\theta_6)^\dagger \sigma_3 R_X(\theta_6) R_Y(\theta_5) R_X(\theta_4) \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_2] \right)^2 \quad (103)$$

$$= \frac{1}{16} \sum_{j \in \{1,2,3\}} \sum_{m \in \{2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_5)} \left(\frac{\partial}{\partial \theta_k} \text{Tr}[R_X(\theta_4)^\dagger R_Y(\theta_5)^\dagger \sigma_m R_Y(\theta_5) R_X(\theta_4) \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_2] \right)^2 \quad (104)$$

$$= \frac{1}{8} \sum_{j \in \{1,2,3\}} \sum_{m \in \{1,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_4)} \text{Tr}[R_X(\theta_4)^\dagger \sigma_m R_X(\theta_4) \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_2]^2 \quad (105)$$

$$\geq \frac{1}{16} \sum_{j, m \in \{1,2,3\}} \mathbb{E}_{(\gamma, \theta_1, \dots, \theta_3)} \text{Tr}[\sigma_m \otimes \sigma_j \otimes \sigma_{0^{N-2}} \rho_2]^2 \quad (106)$$

$$\geq \frac{1}{32} \sum_{j \in \{1,2,3\}} \text{Tr}[\sigma_j \otimes \sigma_{0^{N-1}} \rho_{\text{in}}]^2 \quad (107)$$

Eq.(102) is derived by using Eqs.(64)-(66). Eq.(103) is obtained by using the definition (58). Eq.(104) is obtained by using Lemma A.2. Eq.(105) is obtained by using Lemma A.3. Eq.(106) is obtained by using Lemma A.2. Eq.(107) is derived by using Eqs.(76)-(82). Similar to the case $k = 8$, Eq.(107) could be obtained for $\mathbf{G}_2 \in \{\sigma_0 \otimes |\mathbf{0}\rangle\langle\mathbf{0}| + \sigma_1 \otimes |\mathbf{1}\rangle\langle\mathbf{1}|, |\mathbf{0}\rangle\langle\mathbf{0}| \otimes \sigma_0 + |\mathbf{1}\rangle\langle\mathbf{1}| \otimes \sigma_3\}$ as well by a derivation similar to Eqs.(102)-(107). Now we have proved Theorem E.1.

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